

QIXIA LUO

The University of British Columbia
Department of Mathematics
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EDUCATION

The University of British Columbia, PhD Applied Mathematics

September, 2025 - Present, Vancouver, BC, CA

GPA: 93.5/100

- Advisor: Prof. Elina Robeva
- Research topics: Algebraic statistics; causal inference; data science

Queen's University, MSc Mathematics

September, 2023 - July, 2025, Kingston, ON, CA

GPA: 4.30/4.30

- Advisor: Prof. James A. Mingo
- Research topics: Random matrix theory; free probability
- Thesis: The north pole problem and recent developments in infinitesimal free probability

University of Toronto, HBSc Mathematics & Statistics

September, 2017 - June, 2022, Toronto, ON, CA

GPA: 3.95/4.00

- Degree: Honours Bachelor of Science with High Distinction
- Program: Mathematics & Its Applications Specialist (Probability/Statistics)

RESEARCH INTERESTS

I am interested in the mathematical understanding of statistical and machine learning phenomena. I am interested in establishing frameworks for describing general statistical and machine learning problems from the perspectives of algebraic geometry, multilinear algebra, and combinatorics. Currently I am working in the areas of algebraic statistics and causal discovery.

TEACHING & MENTORING

Summer Undergraduate Reading Program Mentor, The University of British Columbia

June - July, 2026, Vancouver, BC, CA

- I will mentor a small group of undergraduate students on the project *Finite Frames and its Applications* over two months; I will design and structure all learning material and supervise final projects

Tutor, TutorBright

January, 2020 - June, 2022, Toronto, ON, CA

- Provided one-on-one tutoring for high school students in all mathematics subjects

Tutor, Self-employed

September, 2023 - present, Remote

- Provided one-on-one tutoring and mentoring for high school students in all mathematics subjects
- Accumulated more than 300 hours of tutoring and individualized lesson planning

EMPLOYMENT

Teaching Assistant, The University of British Columbia

September, 2025 - present, Vancouver, BC, CA

- Graded homework assignments and midterms for Differential Calculus with Applications (MATH 100), Differential Calculus and Multivariable Calculus (MATH 253), Introduction to Discrete Mathematics (MATH 341)
- Tutored students in the Math Learning Center (MLC)
- Invigilated exams

Teaching Assistant, Queen's University

September, 2023 - August, 2025, Kingston, ON, CA

- Graded homework and exams for linear algebra, integral and differential calculus, complex analysis, vector analysis
- Facilitated tutorials for integral and differential calculus
- Proctored exams

Intern, International Air Transport Association (IATA)

July, 2022 - April, 2023, Montréal, QC, CA

- Conducted business data analytics and created reports
- Supported the global operations for IATA Training in a large international team setting

SPEAKING EXPERIENCE

Reading Group Speaker, The University of British Columbia

- Introduction to Free Probability and Random Matrices; *one talk, 50 minutes, Learning Seminar on Tensors and Causal Inference*
- Representation Theory of $GL(\mathbb{C}^n)$; *two talks, 50 minutes each, Reading Group on Algorithms in Invariant Theory*

Departmental Seminar Speaker, Queen's University

- Stieltjes Transform and its Inversion; *two talks, 50 minutes each, Random Matrix Theory Seminar*
- Joint Infinitesimal Free Cumulants of Certain Unitarily Invariant Matrix Ensembles and their Transposes; *one talk, 50 minutes, Random Matrix Theory Seminar*
- Multiplicative Linear Functionals on Unital Banach Algebras; *two talks, 50 minutes each, Operator Algebras Seminar*
- Computing Maximal Ideal Spaces on Unital Banach Algebras; *two talks, 50 minutes each, Operator Algebras Seminar*
- Commutative C^* -Algebras; *one talk, 50 minutes, Operator Algebras Seminar*
- The Johnson-Lindenstrauss Lemma; *one talk, 50 minutes, Graduate Seminar*

CONFERENCES & WORKSHOPS

Attendant & Speaker, Women in Mathematics of Data Science Workshop

August 4 -14, 2026, Vancouver, BC, CA

Speaker, International Congress on Mathematical Software

July 20 - 23, 2026, Waterloo, ON, CA

Speaker, Society for Industrial and Applied Mathematics Annual Meeting 2026 (SIAM AN26)

July 6 - 10, 2026, Cleveland, OH, USA

Attendant, Upper Bound AI Conference

May 19 - 22, 2026, Edmonton, AB, CA

Attendant, UBC SIAM Chapter Applied Mathematics Meeting

October 25, 2025, Vancouver, BC, CA

Attendant, Ontario Graduate Mathematics Conference (OGMC)

May 30 - June 1, 2025, Toronto, ON, CA

Attendant, AI4GoodLab

May, 2020 - June, 2020, Edmonton, AB, CA (remote due to pandemic)

- 7 weeks full-time workshop with a focus on the social implications of AI
- Amongst a small cohort of women and gender-diverse individuals selected across Canada to participate

GRADUATE AWARDS & SCHOLARSHIPS

- Four Year Doctoral Fellowship, *The University of British Columbia*, \$72,800 total over 4 years
- Tuition Award, *The University of British Columbia*, full tuition coverage over 4 years
- President's Academic Excellence Initiative PhD Award, \$840 per annum, *The University of British Columbia*
- The Senator Frank Carrel Fellowship, *Queen's University*, \$30,000 total over 2 years

UNDERGRADUATE AWARDS & SCHOLARSHIPS

- Howard Ferguson Scholarship, *University of Toronto*, \$12,000 total over 4 years
- The Coxeter Scholarship In Mathematics, *University of Toronto*, \$1480, one time recipient
- Galois Fund, *University of Toronto*, \$750 total, two times recipient
- C L Burton Open Scholarship, *University of Toronto*, \$400, one time recipient
- Dr. James A. & Connie P. Dickson Scholarship In Science & Mathematics, *University of Toronto*, \$400, one time recipient
- Dean's Scholar, *University of Toronto*, 2018 Summer, 2020 - 2022 Winters
- Hugh G. White Memorial Scholarship, *Actuarial Foundation of Canada*, \$1200, one time recipient

VOLUNTEERING & SERVICES

- Application evaluation committee for the 2026 AI4GoodLab cohort
- Grader for the 2025 Canadian Open Mathematics Contest (COMC)

COMPUTING SKILLS

Familiar with Python, R, JavaScript, MATLAB, Julia

MISCELLANEOUS

- Citizenship: Canada
- First name pronunciation: /tʃiːʃə/ or *chee-shuh*
- This *curriculum vitae* is up-to-date as of *June 1, 2026*